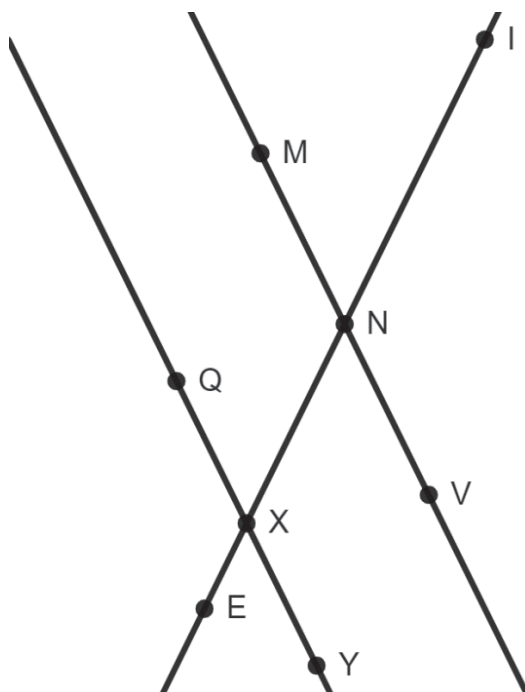


Lines $\overline{MV}$, $\overline{QY}$, and $\overline{EI}$ are shown.

$\overline{MV} \parallel \overline{QY}$, $\angle MNX = 130^\circ$



Identify the relationship between each pair of angles.

Angles	Congruent or Supplementary?
$\angle MNX$ and $\angle QXN$	1.
$\angle MNX$ and $\angle MNI$	2.
$\angle MNX$ and $\angle VNI$	3.
$\angle MNX$ and $\angle QXE$	4.

Complete the table according to the diagram.

Angle	Angle Measure	Relationship with $\angle EXY$
$\angle VNI$	5.	6.
$\angle YXN$	7.	8.

9. If $\angle MNX = (10a)^\circ$ and $\angle QXN = (4a - 2)^\circ$, what is the value of a ?

10. If $\angle EXY = (4w - 6)^\circ$ and $\angle VNI = (9w + 4)^\circ$, what is the value of w ?